

brain-drain bay card: ASM1153E USB-SATA bridge, switched 12 V / 5 V, 22-pin SATA receptacle

bridge

ASM1153E USB 3 to SATA bridge, 30 MHz crystal, core switcher, AC coupling

File: sheets/bridge.kicad_sch

bay-switch

Switched 12 V and 5 V for the drive: P-FET high-side switches with soft-start, PTC fuses

File: sheets/bay-switch.kicad_sch

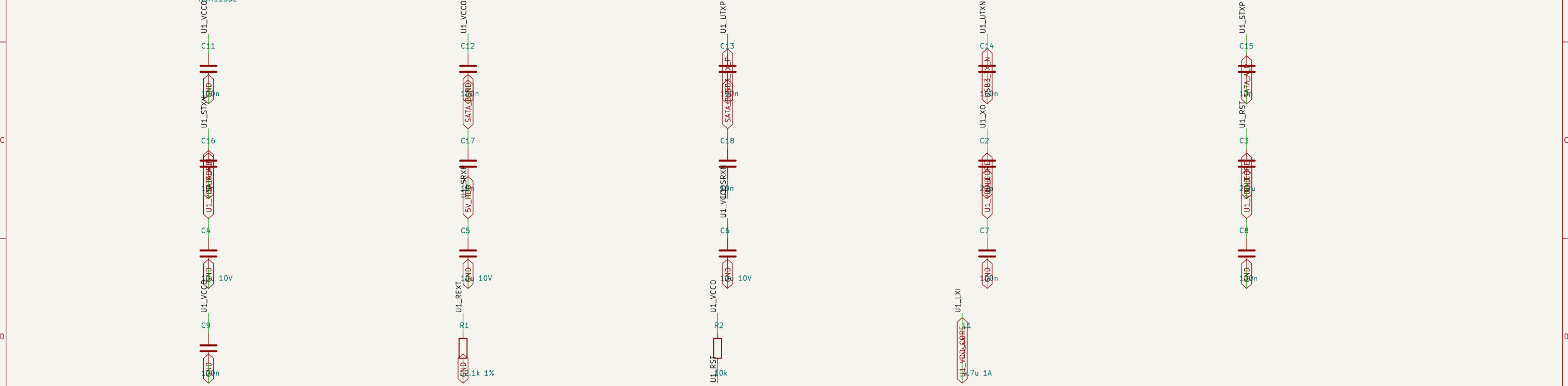
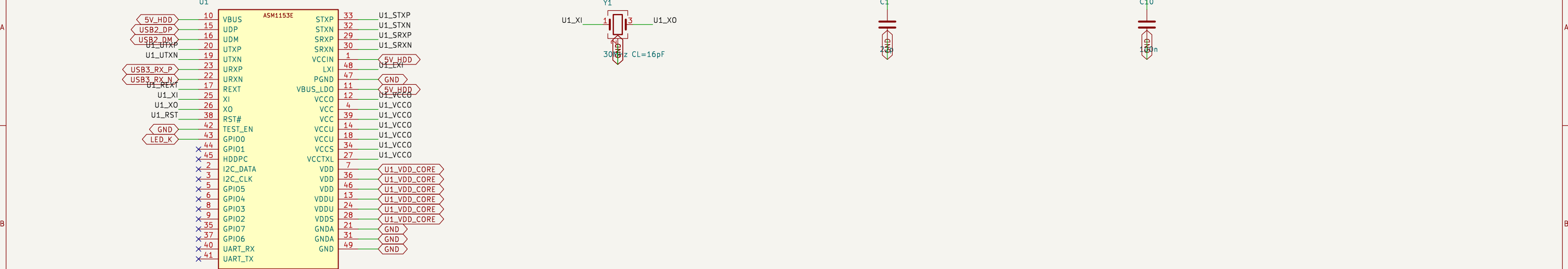
edge

Card edge to the brain (PCIe x1 fingers, bay-card pinout) and the SATA 22-pin receptacle

File: sheets/edge.kicad_sch

brain-drain			
Sheet: /			
File: bay-card.kicad_sch			
Title: brain-drain bay card: ASM1153E USB-SATA bridge, switched 12 V / 5			
Size: A3	Date:		Rev: v0-generated
KiCad E.D.A. 9.0.8	Id: 1/4		

Bridge: ASM1155E USB 3 to SATA Bridge, 50 MHz Crystal, Core switcher, AC coupling	5	6	7	8
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Clock straps GPIO3/GPIO7 at their internal pull-up default (11) = 30 MHz crystal. GPIO6 default = I2C mode, no SPI ROM. Bay activity LED assumed on GPIO0 (verify on the bench); it sinks the brain's LED through the edge.

Sheet: /bridge/
File: bridge.kicad_sch

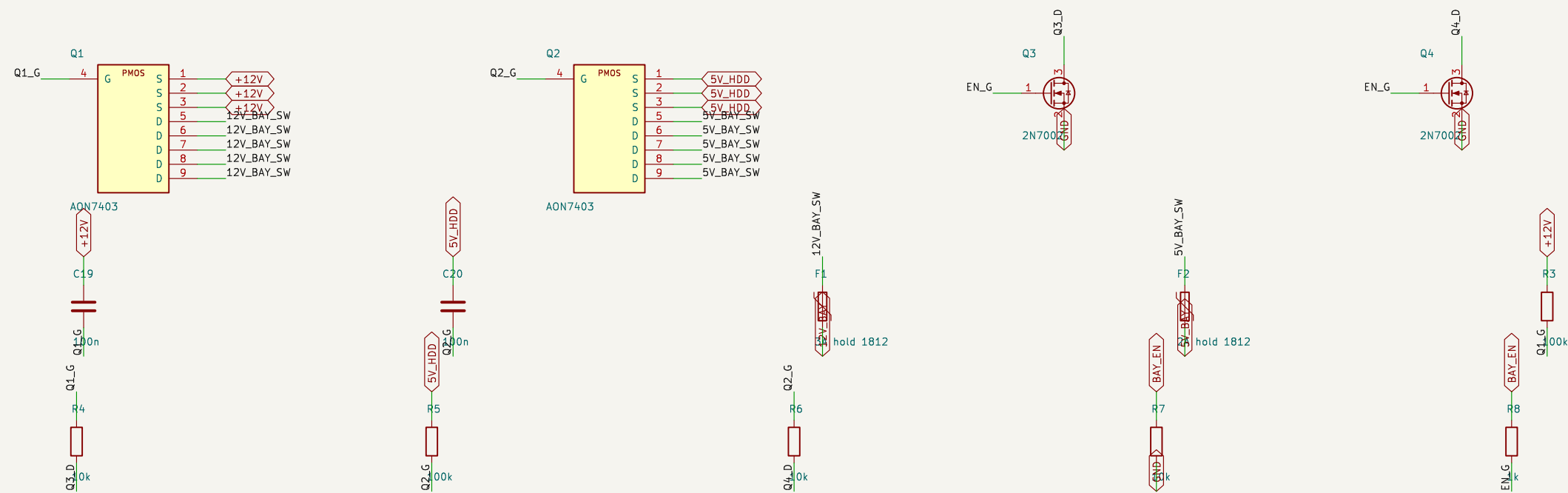
Size: A3	Date:	Rev: v0-generated
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Date:

Rev: v0-generated

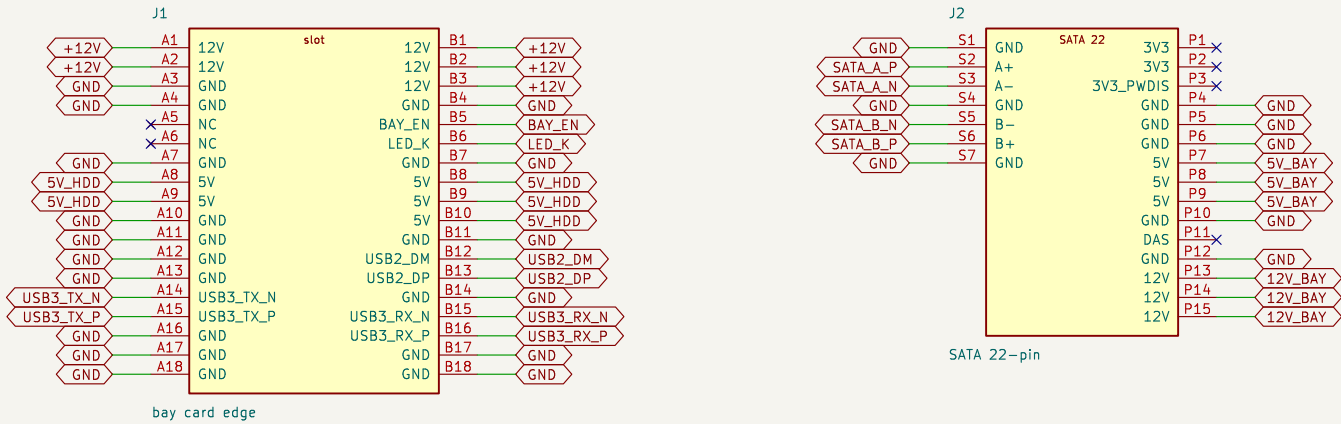
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bay-switch: Switched 12 V and 5 V for the drive: P-FET high-side switches with soft-start, PTC fuses



P-FET high-side switch: 100k gate pull-up holds it off; the 2N7002 pulls the gate down through 10k with 100n gate-source, ~1 ms ramp so the drive's bulk caps do not trip the fuse. BAY_EN pull-down keeps the bay off through

brain-drain					
Sheet: /bay-switch/ File: bay-switch.kicad_sch					
<i>Title: bay-card - bay-switch</i>					
Size: A3		Date:		Rev: v0-generated	
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Fingers: Connector_PCBEde:BUS_PCIexpress_x1 on the card's bottom edge (chamfer the edge, ENIG is fine for a few insertions). Pinout in symgen.SLOT_PINS.
SATA P1-P3 (3.3 V) are not connected on purpose: P3 is PWDIS on SATA 3.3 drives and a 3.3 V supply there keeps them from spinning up.